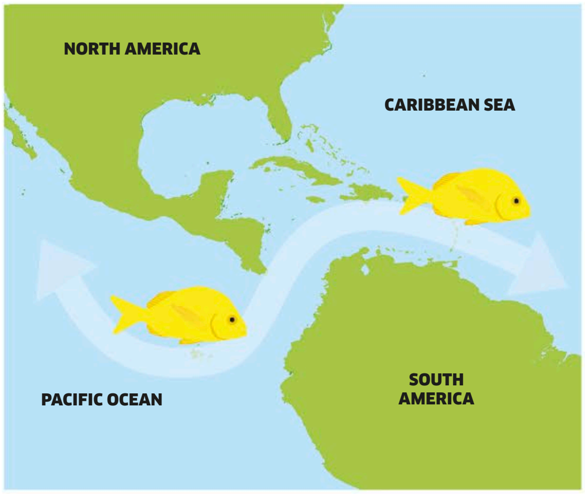
Africa's **Lake Malawi** contains more than **500 species of cichlid fishes** that have **evolved from a single ancestral fish** within the last million years.

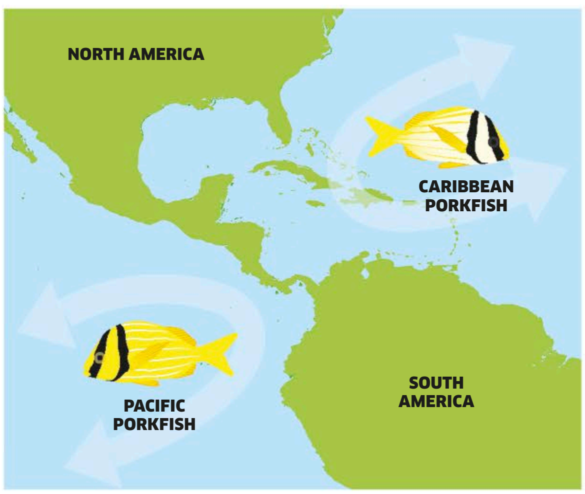
50 years is all the time it takes for **some infectious bacteria** to **evolve resistance** to antibiotic drugs.

How new species emerge

Over a long period of evolution, varieties of animals can end up becoming so different that they turn into entirely new species—a process called speciation. This usually happens when groups evolve differences that stop them from breeding outside their group, especially when their surroundings change so dramatically that they become physically separated from others.



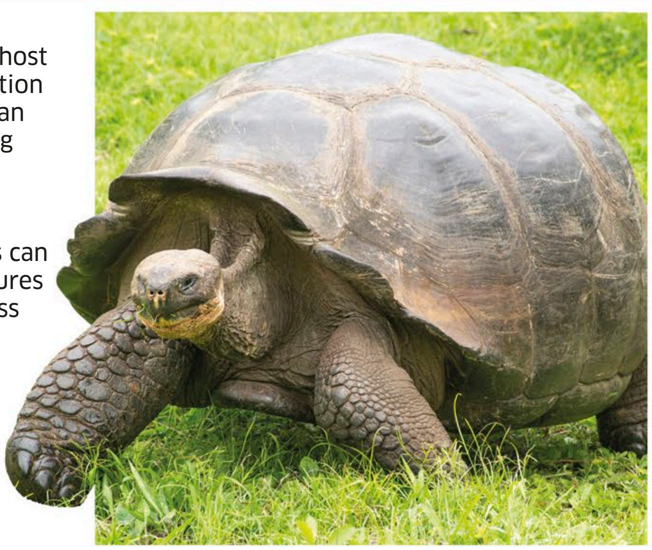
1 Ancestral species Five million years ago, before North and South America were joined, a broad sea channel swept between the Pacific Ocean in the west and the Caribbean in the east. Marine animals, such as the reef-dwelling porkfish, could easily mix with one another in the open waters. Porkfish from western and eastern populations had similar characteristics and all of them could breed together, so they all belonged to the same species.



2 Modern species The shifting of Earth's crust caused North and South America to collide nearly 3 million years ago. This cut off the sea channel, isolating populations of porkfish on either side of Central America. Since then, the two populations have evolved so differently that they can no longer breed with each other. Although they still share a common ancestor, today the whiter Caribbean porkfish and the yellower Pacific porkfish are different species.

Evolution on islands

Isolated islands often play host to the most dramatic evolution of all. Animals and plants can only reach them by crossing vast expanses of water, and—once there—evolve quickly in the new and separate environment. This can lead to some unusual creatures developing—such as flightless birds and giant tortoises.



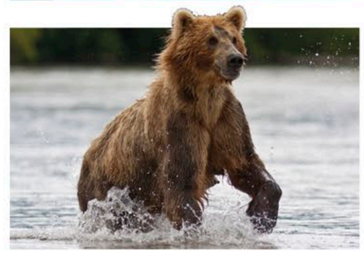
Tortoise travels
The famous giant tortoises unique to the Galápagos Islands are descended from smaller tortoises that floated there from nearby South America.

Out of all the reptiles and land mammals of the Galápagos Islands, 97 percent are found nowhere else in the world.

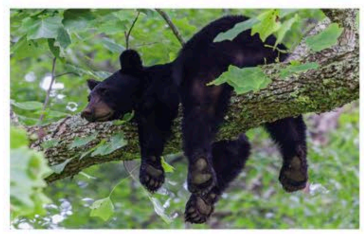
Adaptation

Living things that survive the grueling process of natural selection are left with characteristics that make them best suited to their surroundings. This can be seen in groups of closely related species that live in very different habitats—such as these seven species of bears.

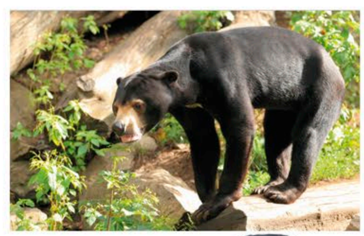
Polar bear
The biggest, most carnivorous species of bear is adapted to the icy Arctic habitat. It lives on fat-rich seal meat and is protected from the bitter cold by a thick fur coat.



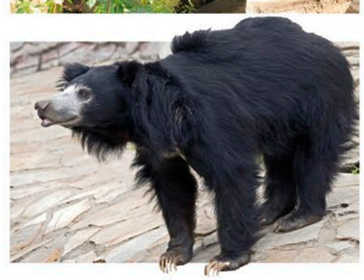
Brown bear
The closest relative of the polar bear lives further south in cool forests and grassland. As well as preying on animals, it supplements its diet with berries and shoots.



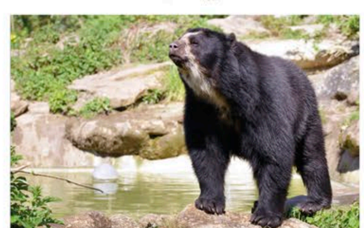
Black bear
The North American black bear is the most omnivorous species of bear, eating equal amounts of animal and plant matter. This smaller, nimbler bear can climb trees to get food.



Sun bear
The smallest bear lives in tropical Asia and has a thin coat of fur to prevent it from overheating. It has a very sweet tooth and extracts honey from bee hives with its long tongue.



Sloth bear
This shaggy-coated bear from India is adapted to eat insects. It has poorly developed teeth and, instead, relies on long claws and a long lower lip to obtain and eat its prey.



Spectacled bear
The only bear in South America has a short muzzle and teeth adapted for grinding tough plants. It feeds mainly on leaves, tree bark, and fruit, only occasionally eating meat.

Giant panda
The strangest bear of all comes from the cool mountain forests of China. It is almost entirely vegetarian, with paws designed for grasping tough bamboo shoots.